

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester 3)
DSA

Practical 7

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Class : SYBSC IT	Batch : 1st
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Practical 7A

Hash Table:

Write a program to:

a. Implement a hash table with separate chaining for collision handling.

Code :

```
size = 5
```

```
table = [[] for _ in range(size)]
```

```
def insert(key, value):
```

```
    table[(key % size)].append((key, value))
```

```
insert(10, "Raj")
```

```
insert(15, "Amit")
```

```
insert(20, "Kaushal")
```

```
insert(25, "Jay")
```

```
print(table)
```

```
size = 5
```

```
table = [[] for _ in range(size)]
```

```
def insert(key, value):
```

```
    table[(key % size)].append((key, value))
```

```
insert(10, "Raj")
```

```
insert(15, "Amit")
```

```
insert(20, "Kaushal")
```

```
insert(25, "Jay")
```

```
print(table)
```

```
|
```

Output :

```
===== RESTART: C:/Users/IT-47/Desktop/Raj_45/DSA 7.py =====  
> [[(10, 'Raj'), (15, 'Amit'), (20, 'Kaushal'), (25, 'Jay')], [], [], [], []]  
|
```

Practical 7B

b. Store and retrieve data from the hash table.

Code :

```
size = 5
```

```
table = [[] for _ in range(size)]
```

```
def insert(key, value):
```

```
    table[(key % size)].append((key, value))
```

```
def search(key):
```

```
    for k, v in table[key % size]:
```

```
        if k == key:
```

```
            return v
```

```
    return "Not Found"
```

```
insert(10, "Raj")
```

```
insert(15, "Amit")
```

```
insert(20, "Kaushal")
```

```
insert(25, "Jay")
```

```
print(table)
```

```
key = int(input("Enter Key"))
```

```
print(search(key))
```

```

size = 5
table = [[] for _ in range(size)]

def insert(key, value):
    table[(key % size)].append((key, value))

def search(key):
    for k, v in table[key % size]:
        if k == key:
            return v
    return "Not Found"

insert(10, "Raj")
insert(15, "Amit")
insert(20, "Kaushal")
insert(25, "Jay")

print(table)

key = int(input("Enter Key"))
print(search(key))

```

Output :

```

===== RESTART: C:/Users/IT-47/Desktop/Raj_45/DSA 7.py =====
[[ (10, 'Raj'), (15, 'Amit'), (20, 'Kaushal'), (25, 'Jay') ], [], [], [], []]
Enter Key10
Raj

```

```

===== RESTART: C:/Users/IT-47/Desktop/Raj_45/DSA 7.py =====
[[ (10, 'Raj'), (15, 'Amit'), (20, 'Kaushal'), (25, 'Jay') ], [], [], [], []]
Enter Key1111
Not Found

```